

LAB#4

Half Adder & Full Adder

x	y	Sum	Count
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

SOP:-

$$\bar{x}y + x\bar{y} \rightarrow \text{sum}$$

$$\text{sum} = x \oplus y$$

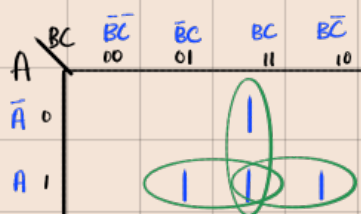
A	B	C	Sum	Carry out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$\rightarrow \bar{A}\bar{B}C$

$\rightarrow \bar{A}B\bar{C}$

$\rightarrow A\bar{B}\bar{C}$

Full Adder has 3 input
so $2^3 = 8$



$$\text{Count} = AC + BC + AB$$

$$\text{sum} = A \oplus B \oplus C$$

$$\text{Count} = AC + BC + AB$$

$$\text{sum} = A \oplus B \oplus C$$